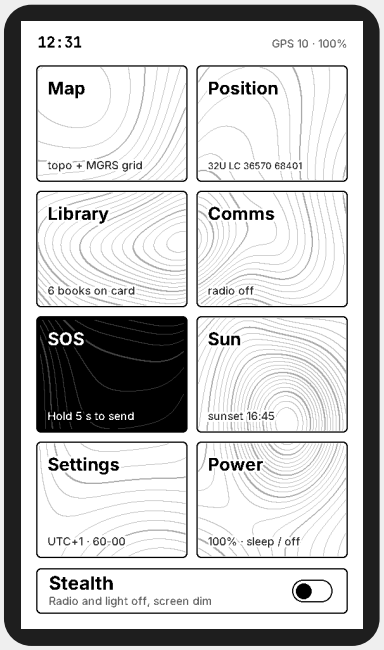
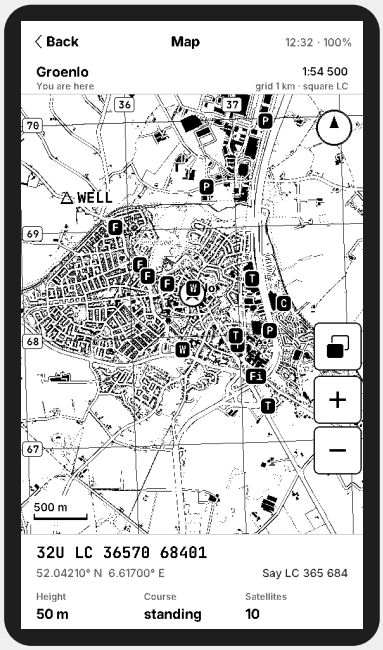
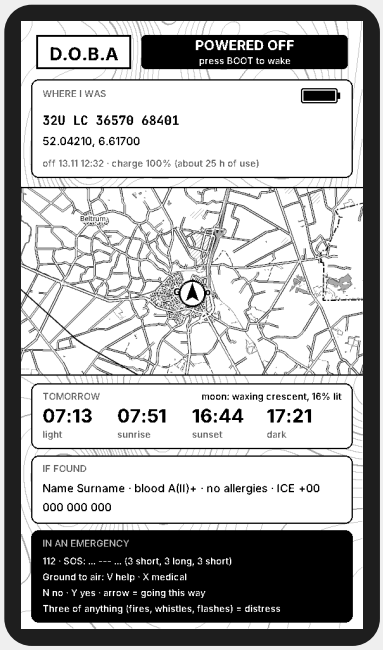
D.O.B.A

Durable Offline Backcountry Assistant

An e-ink pocket device that keeps working when the internet, the phone network and the power grid don't: topographic maps with a military grid, an offline library, text messaging over LoRa, and a paper card when it is switched off.







Firmware + data pipeline for the LilyGo T5 E-Paper S3 Pro · English / Українська
github.com/festchenko/doba

What it does

Maps — Vector topographic tiles drawn for 4-grey e-ink: roads, buildings, forests, water, contour lines and heights. North-up or course-up, from street level to a whole country. Most of Europe plus Ukraine, built region by region from OpenStreetMap.

Navigation — MGRS position to 1 m, latitude/longitude, magnetic declination, azimuth in mils (6000 or 6400) and degrees. A measuring cross gives distance and compass azimuth from you or from point A, and the ground height or water depth under it.

Points & places — Save points from GPS, from the map or typed as MGRS, then 'go here' with distance and azimuth. Nearby: hospitals, doctors, pharmacies, water, food shops, farms, shelters, bunkers and underground parking, offline.

Library — Kiwix archives converted to a light format: Wikipedia (EN, UK) with pictures, WikiMed, iFixit, Appropedia, Stack Exchange, Wikisource and Gutenberg classics, Wikivoyage, Wiktionary, field manuals and your own PDF/EPUB. Topic pages and three emergency buttons.

Reader — Serif text with Cyrillic hyphenation, pictures and captions, contents, links, text size. Long books reopen where you stopped, even after a restart.

Comms — LoRa SX1262, Meshtastic-compatible channels and direct messages, nodes with distance and bearing, quick phrases, an EN/UA keyboard. SOS repeats your MGRS position every 5 minutes.

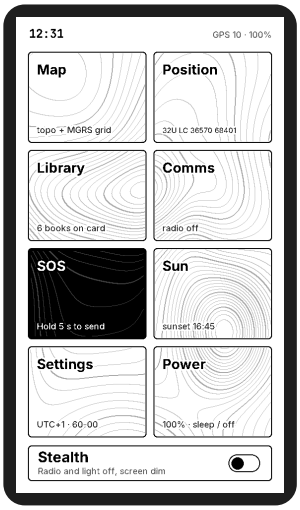
Stealth — One switch: radio silent or listen-only, light off, screen dim. Bluetooth and Wi-Fi are off by default.

Sun & battery — Sunrise, sunset, twilight, daylight left, moon phase. Battery: measured current, time left at this pace, and where the charge goes.

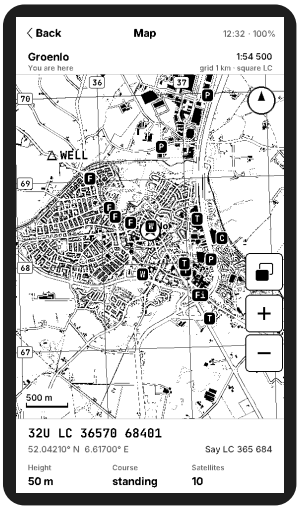
Power-off card — Switched off, the battery is disconnected (ship mode) and the glass keeps: where you were, when, the charge, tomorrow's light, an 'if found' card and an emergency crib.

Screens

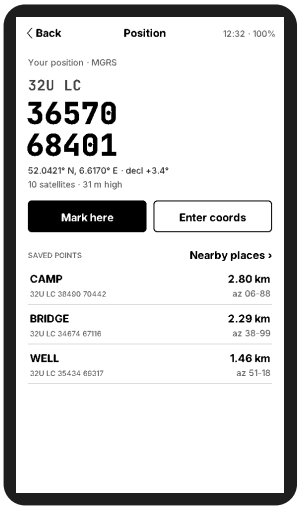
Rendered by the firmware itself on the desktop simulator, with a real map of Groenlo (NL).



Home



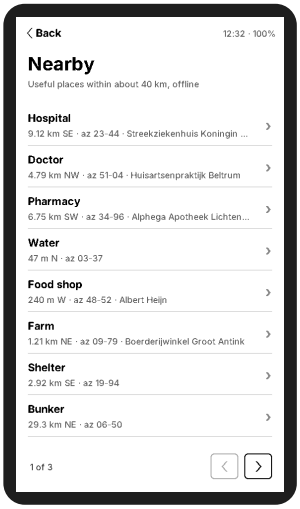
Topo map + MGRS



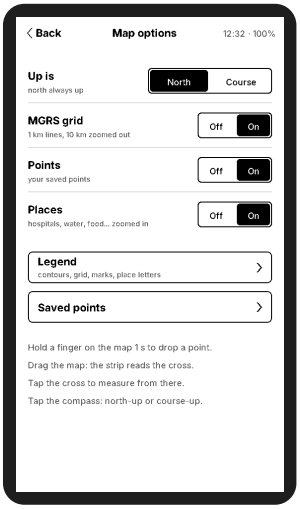
Position & points



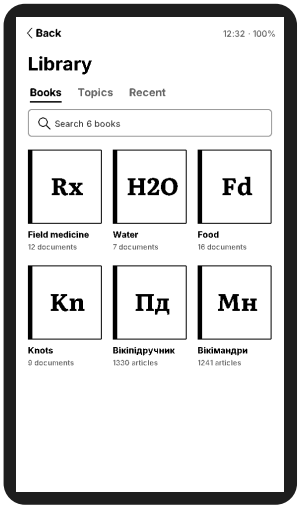
Go to a point



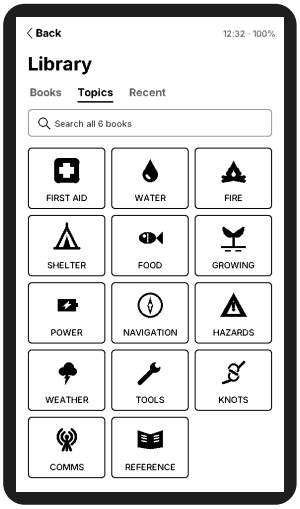
Places nearby



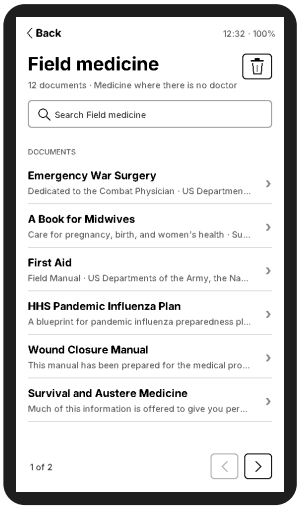
Map options



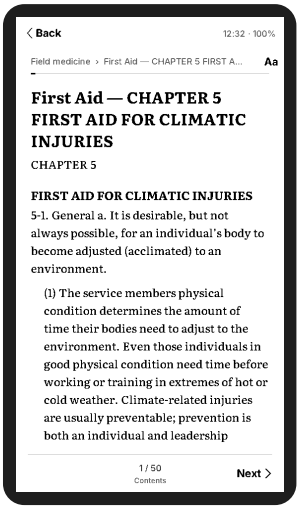
The library shelf



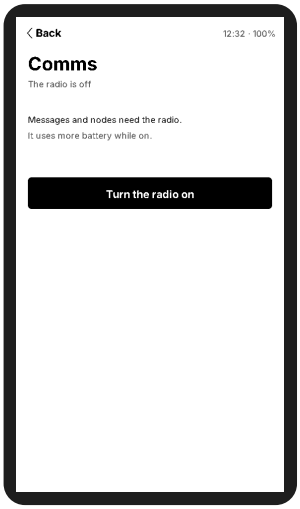
Library topics



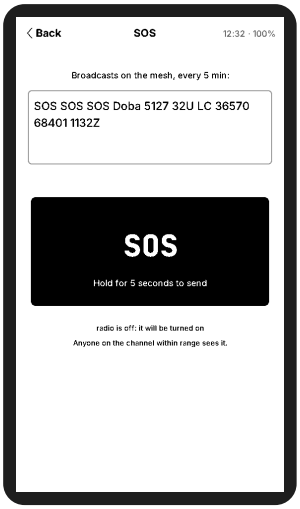
A book's own page



Reading



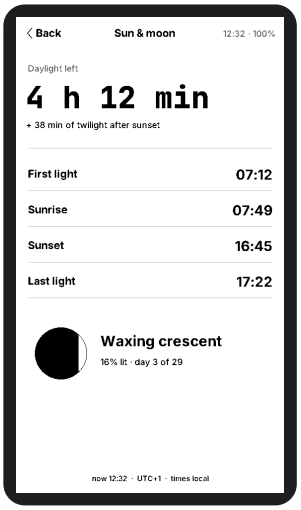
Comms (LoRa)



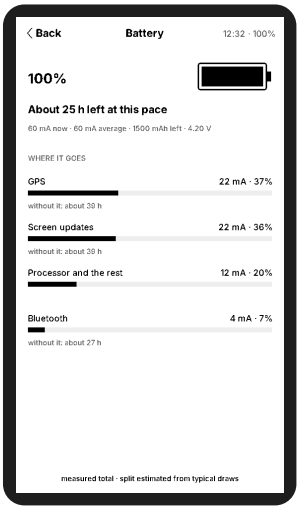
SOS

Screens

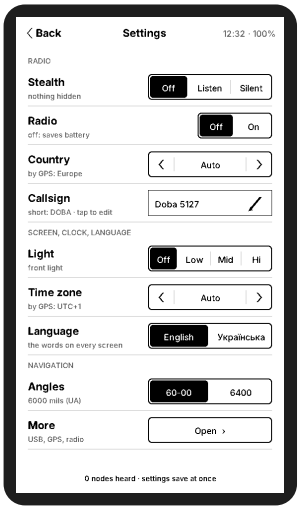
Rendered by the firmware itself on the desktop simulator, with a real map of Groenlo (NL).



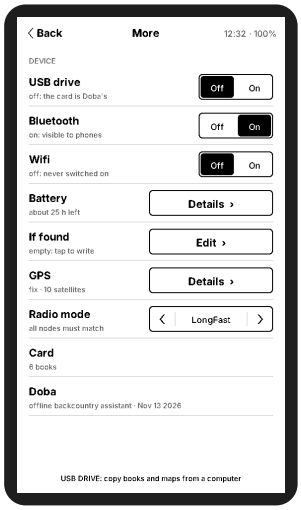
Sun and moon



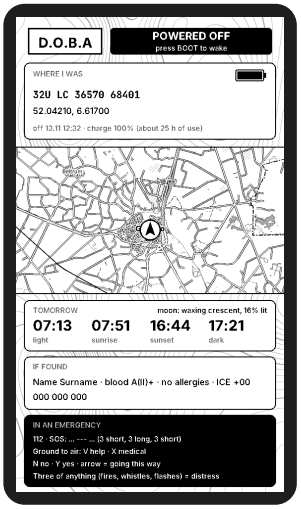
Battery



Settings



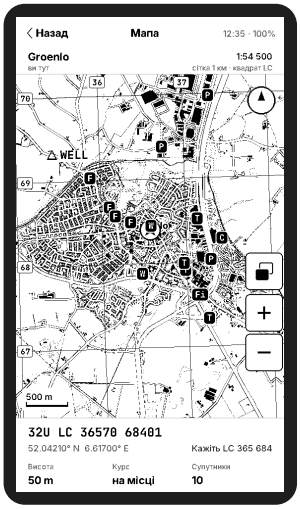
Device



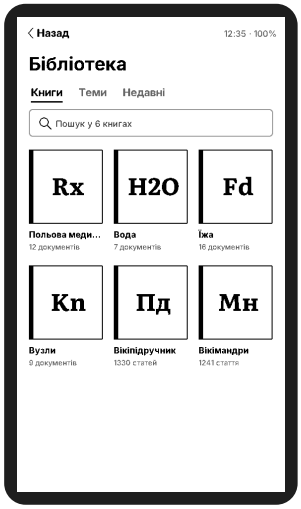
The power-off card



Українською



Мапа



Бібліотека



Екран вимкнення

Hardware & where to get it

LilyGo T5 E-Paper S3 Pro (H752-01)

ESP32-S3, 16 MB flash, 8 MB PSRAM, 4.7-inch 960x540 touch e-ink, GPS, SX1262 LoRa, 1500 mAh battery, RTC. Choose the version with GPS and the LoRa band of your region (868 MHz in Europe and Ukraine).

→ lilygo.cc/products/t5-e-paper-s3-pro
→ tindie.com/products/lilygo/lilygo-t5-e-paper-s3-pro
→ LilyGo's official AliExpress store

microSD card, 256 GB to 1 TB

Formatted FAT32 by the scripts. Any A1/A2 card works: the device reads over SPI, so the speed class only shortens the copy from the computer.

Optional: a second Meshtastic node

To message with Doba or as a relay, for example a Heltec V3 at 868 MHz.

Getting started

1. Flash: python3 -m esptool --chip esp32s3 write-flash 0x0 release/doba-full.bin
2. Build maps per region from OpenStreetMap: maps/make-region.sh <region>
3. Build the library from Kiwix archives: library/fetch_zims.sh, then 'convert'
4. Add your own books: library/docs2doba.py ~/Books .../lib/my_books
5. Format the card and copy everything: tools/prepare-sd-mac.sh format diskN, then 'all'
6. Buttons: BOOT short = home, hold = power menu; side IO48 = light; POWER = switch on; RST = restart

Credits & support

Firmware is a fork of OpenTrailPaper by Raemond Bergstrom-Wood (Apache-2.0). Map data © OpenStreetMap contributors (ODbL). Library content comes from Kiwix archives under their own licenses and is not part of the repository. Meshtastic is a trademark of Meshtastic LLC; Doba speaks its open protocol and is not affiliated. Doba is not a certified safety or medical device.

If Doba is useful to you, support its development: github.com/sponsors/festchenko